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Research Paper

Effects of Sanitation Practices on Microbial Dynamics in Meat Processing Environment

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ABSTRACT

This study investigated the effects of a multistage sanitation process on the microbial populations associated with conveyor belts, drains, and air within a large commercial beef processing facility. Total aerobic counts in samples from conveyor belts, drain, and air increased after a pressurized warm water wash ($P < 0.05$), decreased after foaming and degreasing ($P < 0.05$), and were not affected ($P > 0.05$) by application of peracetic acid (500 ppm) or quat−based (200 ppm) no-rinse sanitizers at recommended in-use concentrations with a 5−min contact time. *Enterobacteriaceae* and coliforms counts on conveyor belts and drains largely followed the same pattern as those of total aerobic counts. However, the *Enterobacteriaceae* counts in air samples were below the detection limit (1 CFU/100 L). *Escherichia coli* were not recovered from belts before cleaning or after sanitation, but were detected only sporadically during sanitation. In drain samples where *E. coli* were recovered, counts were not affected by cleaning or sanitation steps. Sequencing results revealed that the microbial composition varied by different sampling trips. Overall, *Acinetobacter* was predominant throughout the sanitation process in conveyor belt, drain, and air samples, with overall relative abundance of 46.06%, 51.18%, and 55.83%, respectively. Prediction models based on sequencing data indicated that the drain surface was a significant contributor to the initial microbiota on conveyor belts, but was replaced by air at the step of pressurized water washing.

A routine sanitation program is mandatory in meat processing facilities in North America, in particular US and Canada, to ensure hygiene condition of the processing environment, including equipment surfaces (Marriott et al., 2018). The standard sanitation processes employed by most meat processing plants involve three main stages: removal of visible soil, cleaning and degreasing with detergent, and applying chemical disinfectants (Wang et al., 2018). These procedures are designed to remove meat debris, fat, oil, and grease deposits as well as to reduce the microbial load on food contact surfaces (Li et al., 2019). Despite the implementation of these standardized sanitation protocols, microbial contamination of meat products during processing remains a challenge, and this affects microbial safety and shortens the expected shelf life of meat products. Recently, Zwirzitz and coworkers identified potential microbial transmission routes between pork carcasses and cuts from various processing sites (e.g., equipment, staff, and infrastructure of the facility), based on high-throughput sequencing and modeling with bioinformatics tools (Zwirzitz et al., 2020). In addition, *Pseudomonas* and *Psychrobacter* were shown to survive cleaning and disinfection, likely due to their ability to form biofilms and resistance to biocides (Wagner et al., 2020).

Aerosolized bacteria generated by actions such as hide pulling during the beef dressing process have been recognized as a source of cross-contamination, for example, Shiga toxin-producing *E. coli* and *Salmonella* were frequently detected in the air near the hide pulling station (Beck et al., 2019; Mohammad et al., 2021). For the soil-removal stage of the facility cleaning process, a pressurized warm water wash is applied (Wang et al., 2018). Even though it facilitates the removal of food debris, thereby reducing visible contamination on surfaces, it has to be recognized that pressurized water wash can generate aerosols, i.e. microparticles, as motions of hide pulling may do. These microparticles range in size from a few micrometers to

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100 μm, and once aerosolized, can travel significant distances and be deposited on downward surfaces at different times depending on the size of the droplets (Lutgring et al., 1997). Sheehan and Giranda reported higher bacterial counts during concurrent production and sanitation that used high-pressure spraying, compared to production alone, in the air of a food processing facility, suggesting the potential for pressurized water wash to translocate bacteria (Sheehan & Giranda, 1994). However, to date, limited information is available regarding to what extent do bioaerosols generated during cleaning contribute to cross-contamination of food-contact surfaces in commercial facilities.

Federally inspected meat processing plants in Canada are large and complex. For the fabrication facilities where chilled carcasses are made into specific cuts and trimmings, conveyor belts constructed from interlocking plastic pieces are common. Yang and coworkers used genotyping to demonstrate that equipment surfaces in meat processing plants, particularly conveyor belts, can be major sources of *E. coli* on beef cuts (Yang et al., 2015). *E. coli* is a well-established indicator for pathogens of enteric origin that share the same natural habitat. Additionally, floor drains are well-documented sources of diverse and persistent microbial communities including foodborne pathogens and spoilage microbes (Chitlapilly Dass et al., 2020; Fagerlund et al., 2025; Xu et al., 2025). Using a polycarbonate flexi-glass chamber simulating a high−pressure wash in a commercial sanitation process, Saini and coworkers demonstrated that *Listeria innocua*, a surrogate for the pathogen *Listeria monocytogenes*, in floor drains can be translocated to food contact surfaces (Saini et al., 2012). Compared to foodborne pathogens and indicator organisms, little information is available on the effects of sanitation, especially as a function of individual sanitation stages applied at an industrial scale, on the overall microbial load and composition. The background microbiota may pose a threat to food quality and play a role in the persistence of pathogens (Yang, Wang, et al., 2023) and spread of antimicrobial resistance genes (Verraes et al., 2013). Compared to conditions tested in laboratory settings, processing environments differ in the presence of harborage sites, repeated exposure to sublethal concentrations of cleaning agents/sanitizers, and biofilm formation, all of which may shape the microbial ecology in such environments (Alves et al., 2024; Yang, Narvaez-Bravo, et al., 2023).

To address these gaps, a comprehensive investigation was conducted over a period of three months to examine microbial communities throughout various stages of the sanitation process in a commercial beef processing facility. Air samples, conveyor belt surface swabs, and floor drain swabs were collected throughout the sanitation process and analyzed by both conventional culture-based assays and 16S rRNA gene amplicon sequencing.

Materials and methods

Sanitation stages

The fabrication facility of a large beef processing plant, where cattle are processed at rates up to 280 carcasses per hour, was involved in the study (Yang et al., 2012). Cuts fabrication and trimming collection at the facility involved seven conveyor belts running in parallel (Fig. 1). During the sample collection period, the facility used a generic sanitation protocol consisting primarily of three stages: removal of debris, cleaning and degreasing with detergent, and sanitization. Briefly, large pieces of meat were removed from the conveyor belts, and the belts were washed with pressurized warm water (40 to 100 PSI and 55 °C) to remove debris during "precleaning" (Stage 1). Debris on the floor and conveyor belts was also removed by shovelling, and the pressurized water wash was not just limited to the precleaning stage. After precleaning, the belts were sprayed with Powerfoam (PAC510, Fortrex, North Little Rock, AR, USA), a chlorinated alkaline foaming cleaner and degreaser, and followed by rinsing with warm water (Stage 2). Belts were elevated from the base roller during rinsing. An inspection was then performed to make sure all surfaces were

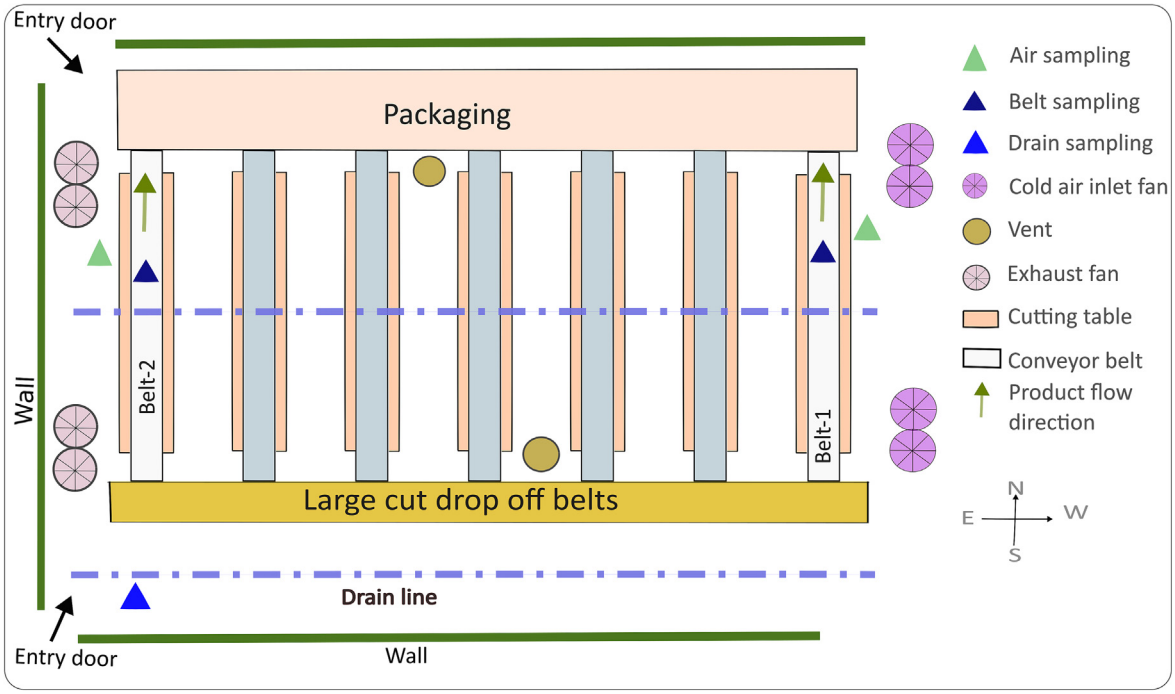


Figure 1. Schematic drawing of sites of sample collection in the fabrication facility. Two belt lines (belt 1 and belt 2) were selected for sampling, as these belt lines were identified as carrying high risks for fat buildup. In addition, belt 1 was located near the cold air inlet and belt 2 was located near the exhaust fan providing an appropriate aerodynamic of plant air circulation during the sanitation process. For conveyor belts and air, samples were collected from both belt lines, whereas for drain, samples were collected only near belt 2.

free of visible contamination. Further foaming, scrubbing, and rinsing were carried out for places that did not pass the inspection. Subsequently, the belts were sprayed with sanitizers in a two-stage manner: targeted spray using a handheld spray gun of areas of concern, followed by spraying of all belt surfaces (Stage 3). Sanitizers used included no rinse peracetic acid sanitizer (CC-610, Fortrex, North Little Rock, AR, USA) at 500 ppm, lactic acid quat (PAC 641, Fortrex, North Little Rock, AR, USA) at 200 ppm, and neutral quat (PAC 610, Fortrex, North Little Rock, AR, USA) at 200 ppm. One of these sanitizers was used in rotation on different days. During the sanitation process, the belts were allowed to continue to run. The time immediately after completion of the work shift on the fabrication floor and before the sanitation process was defined as Stage 0.

Sample collection

The fabrication facility had seven belt lines in total (Fig. 1), two of which (belt 1 and belt 2) were selected for sampling based on the initial consultation with the plant sanitation team. These belt lines were identified as a high-risk zone for fat buildup. Additionally, belt 1 was located near the cold air inlet, and belt 2 was near the exhaust fan, providing an appropriate location to capture the dynamics of air during sanitation. Samples were collected on three different days at monthly intervals during the routine sanitation process from: the two selected conveyor belts (belt-1 and belt-2), one drain near belt-2, and air within the vicinity of belt 1 and belt 2 (Fig. 1). On each day, samples were collected at four time points: Stage 0–3. Specifically, Stage 0 was immediately after completion of shift work; Stage 1 was after debris removal by dry cleaning and washing with pressurized warm water, and right before the application of the detergent Powerfoam; Stage 2 was after the belt surfaces were cleaned with Powerfoam and thoroughly rinsed with ambient water, and before the application of sanitizer; Stage 3 was within 5 min of the application of sanitizers. On each day, at each stage and time point, four samples were collected from the belt and drain surfaces, along with two air samples.

Belt surface and drain samples

For belt surfaces, four areas each approximately 1000 cm^2 were swabbed at each sampling time using a synthetic sponge (SpeciSponge; VWR Canlab, Edmonton, Alberta, Canada). Sponges were each premoistened with 8 ml of 0.1% peptone water (w/v; Difco, BD, Sparks, MD). The sponge was held from outside of the plastic bag containing the sponge and swabbed back and forth along the moving conveyor to cover the 1000 cm^2 that was required. Different strips of the belt were sampled so to avoid overlapping with a previously sampled portion of the belt. The opening of the drain from which samples were collected restricted direct access to the running fluid due to its small grooved opening. Therefore, for the drain, the adjacent and edge surfaces were swabbed, covering approximately 1000 cm^2 for each sample. Sponge samples were placed on ice upon collection and during transportation. The sample container along with ice was kept in a walk-in cooler (2 °C) until processed within 40 h of collection. In total, 144 sponge samples were collected, with 4 at each sampling point for each site on each of the three sampling trips.

Air samples

For air samples, an AirCup air sampler (ACD210; InnovaPrep, Drexel, MO) was used. The air sampler was equipped with an electrostatic filter to retain bacterial cells. The AirCup was mounted on an adjustable tripod, positioned to allow sample collection at the interception of approximately two feet horizontally away from and one foot above the belt of interest (Fig. 1). This was to collect bioaerosols generated during pressurized water washing of the belts and fabrication facility floor. The sampler was programmed to suck air through the filter in the sampling cup at a constant flow rate of 200 L/min. For each sample, the air sampler was operated for 5 min. Duplicate samples were collected at each sampling time. Immediately after sampling, the filter cup was detached from the sampler and aseptically transferred into a sterile bag. The sample bag was immediately placed on ice upon collection and during transportation, later kept in a walk-in cooler (2 °C) until further processed within 40 h after collection. In total, 48 air samples were collected, with 2 at each sampling point for each site on each of the three sampling trips.

Processing of samples for microbiological analysis

All samples were enumerated for total aerobic bacteria, *Enterobacteriaceae*, coliforms, and *E. coli*. For sponges, 10 ml of 0.1% peptone water was added to each bag containing a sponge, and the bag was pummelled in a stomacher (Seward, Worthing, United Kingdom) for 2 min at high speed (260 rpm). The sponge was squeezed within the bag, and the fluid was transferred into a sterile 15 − ml centrifuge tube. An aliquot of 1 ml of each stomacher fluid was used to prepare 10-fold serial dilutions in 0.1% peptone water. The original fluid and appropriate dilutions were plated onto 3M − Petrifilm plates (3M, St. Paul, MN, USA), which were incubated at 25 °C for 72 h. The remaining original stomacher fluid was centrifuged at 10,000 × *g* for 5 min, and the pellets were archived at −80 °C until further processing for DNA extraction.

For air samples, the filter in each sample unit was removed using a sterile surgical blade and was aseptically transferred into a 50 − ml centrifuge tube containing 1 g of glass beads (500 μm; BioSpec Products, Bartlesville, OK, USA) and 10 ml washing buffer (InnovaPrep, Drexel, MO, USA). The washing buffer was 0.15% Tween 20 and 0.05% Triton X-100 in phosphate buffer (PBS). Centrifuge tubes were manually shaken to wet the filter surfaces and vortexed for 30 sec. Each filter was then transferred into a 10 − ml syringe using sterile forceps. The solution retained by the filter was then ejected. A 1 ml aliquot of the suspension was 10-fold serially diluted and processed for enumeration of total aerobes, *Enterobacteriaceae*, coliforms, and *E. coli* as described before. An additional 1 ml aliquot from each sample was mixed with glycerol and stored at −80 °C. The remaining original suspension was pelleted and archived at −80 °C until future processing for DNA extraction. Due to the low biomass from air samples, all samples were retrospectively enriched. Briefly, an aliquot of 100 μL of the original samples from the glycerol stock was mixed with 5 mL of half-strength brain heart infusion broth and incubated in a shaking incubator at 25 °C for 72 h. After incubation, an aliquot of 1 mL of each enrichment culture was transferred into a 1.5 − mL Eppendorf tube. The cell suspension was pelleted and processed for DNA extraction.

DNA extraction and 16S rRNA amplicon sequencing

Bacterial pellets were subjected to DNA extraction using a Qiagen DNeasy® Blood and Tissue Kit (Germantown, MD, USA), following the manufacturer's instructions for Gram-positive bacteria. Extracted DNA was quantified using a Qubit fluorometer (Life Technologies, CA, USA). All DNA samples were subjected to 16S rRNA gene amplicon sequencing by Genome Quebec using an Illumina MiSeq Platform (2 × 300 cycles; Illumina, Inc., San Diego, CA), targeting the V4 region of 16S rRNA genes (Caporaso et al., 2012).

Sequence quality was assessed using FastQC (Andrews, 2010). Sequenced reads were processed using Quantitative Insight into Microbial Ecology2 (QIIME2; 2023.9.1) (Bolyen et al., 2019). The reads were denoised via the Divisive Amplicon Denoising Algorithm 2 (DADA2) and trimmed with lengths of 240-bp forward and 220-bp reverse reads (Callahan et al., 2016). An amplicon sequence variant (ASV) feature table was then created. The taxonomy classifier based on the SILVA database (release 138.1) was used to assign taxonomies (Quast et al., 2012), and a phylogenetic tree was generated using the FastTree 2 plugin in QIIME2 (Price et al., 2010). Subsequently, QIIME2

artifacts, including the ASV feature table, phylogenetic tree, taxonomic classification, and metadata, were converted into a phyloseq object in R (v4.4.0). Sequencing contaminants were removed using sequenced negative controls of DNA extracted as references by the R package Decontam (v3.20) (Davis et al., 2018). Analyses of diversity were performed using the phyloseq R package (1.48.0) (McMurdie & Holmes, 2013). Specifically, alpha-diversity analyses were based on Observed ASVs and Shannon indices, whereas beta-diversity was calculated based on the Bray-Curtis distance matrices. Analyses of diversity were visualized using ggplot2 R package (v3.5.1). Differentially abundant analyses on taxa were performed using Maaslin2 R package (v1.18.0) using mixed-effects models with Compound Poisson Linear Model, where "sampling batch" was set as a random effect and "cleaning_stage" as a fixed effect (Mallick et al., 2021). To assess the contribution of potential microbial sources, a Bayesian-based source tracking package SourceTracker2 was used (Knights et al., 2011). The belt surface at different cleaning stages was regarded as a sink, while the air and drains at different cleaning stages were regarded as potential sources. Further, to assess the interaction of the microbes between belt, air, and drains, each sample site, regardless of the cleaning stages, was set as sinks, whereas the other two sample sites at different cleaning stages were regarded as potential sources. The contribution of microbes from different sources to the target environment was visualized using ggplot2. Sequences of this project were submitted to the NCBI Sequence Read Archive under BioProject PRJNA1290812.

Statistical analysis

All bacterial counts were transformed to log values. For samples where no bacteria were recovered, 0.5 log below the detection limit was arbitrarily assigned for that particular bacterial type for that sample. An analysis of variance (ANOVA) for log counts was performed in R, one-way ANOVA among sanitation stages for drain samples, and a two-way ANOVA for air and conveyor belt samples to analyze the effects of sanitation stages and belt location, followed by *post hoc* group comparisons using the Sidak test ($P < 0.05$). For alpha-diversity analyses, significance was determined by Kruskal–Wallis followed by Dunn's test. Significant clustering of beta-diversity analysis was determined by the adjusted P value of permutated multivariate analysis of variance using the pairwiseAdonis R package (v0.4.1). Significance was indicated as: * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

Results

Change in the numbers of indicator bacteria during sanitation

Table 1 presents the numbers of aerobes recovered from the belt surface, drain, and air samples at all sanitation stages. All except four samples from belt-2 at Stage 2 and Stage 3 yielded aerobic counts. At Stage 0, the mean aerobic count for belt-1 (3.80 ± 0.53 log CFU/1000 cm^2) and belt-2 (2.76 ± 0.78 log CFU/1000 cm^2) did not statistically differ. For both belts, the aerobic counts increased ($P < 0.05$) to >5 log CFU/1000 cm^2 at Stage 1 and were reduced ($P < 0.05$) by >2 log units at Stage 2. Sanitizer application did not significantly affect the mean number of aerobes on either belt. Even so, at postsanitation, the number of aerobes on belt-2 was 1.5 log lower ($P < 0.05$) compared to that for belt-1. For the drain samples, at Stage 0, the number of aerobes was 7.05 ± 0.68 log CFU/1000 cm^2 and was reduced by 2 log at the end of sanitation. Unlike the belts, the precleaning (Stage 1) did not affect the drain aerobic count. As with the belts, the aerobic count was reduced by 1.7 log units at Stage 2 and was not affected by sanitization. Aerobic counts for air close to the belts were similar at Stage 0, at 2.5 log CFU/1000 L. They followed a similar trend to those for the belts during the sanitation process, as they increased at Stage 1, decreased at Stage 2, and remained unchanged at Stage 3.

Enterobacteriaceae and coliforms counts for the two belts responded to sanitation in a manner similar to the aerobic counts (Table 2). At Stage 3, ≥75% of the samples from both belts did not yield any counts for either bacterial group, at a detection limit of 1 CFU/100 cm^2. For the drain samples, the numbers of *Enterobacteriaceae* and coliforms at equivalent stages were similar. The change in the number of the two

Table 1
Aerobic bacteria recovered from conveyor belts ($n = 12$), drain ($n = 12$), and air ($n = 6$) in a beef fabrication facility

Sample	Sanitation		Aerobes (log_{10} CFU/1000 cm^2)	
Type	Stages[A]	N[B]	Belt-1	Belt-2
Belt	Stage 0	0	3.80 ± 0.53^{b}	2.76 ± 0.78^{ab}
	Stage 1	0	5.39 ± 0.77^{c}	5.72 ± 0.15^{c}
	Stage 2	4 (belt-2)	2.97 ± 1.12^{ab}	2.70 ± 1.95^{ab}
	Stage 3	4 (belt -2)	3.48 ± 0.72^{b}	1.92 ± 1.32^{a}
Drain	Stage 0	0	NA	7.05 ± 0.68^{b}
	Stage 1	0	NA	7.49 ± 0.52^{b}
	Stage 2	0	NA	5.80 ± 1.12^{a}
	Stage 3	0	NA	5.01 ± 0.86^{a}
			Aerobes (log_{10} CFU/1000 L)	
Air	Stage 0	0	2.52 ± 0.26^{a}	2.46 ± 0.66^{a}
	Stage 1	0	3.48 ± 0.20^{bc}	3.74 ± 0.25^{c}
	Stage 2	0	2.21 ± 046^{a}	2.32 ± 0.46^{a}
	Stage 3	0	2.57 ± 0.31^{a}	2.86 ± 0.65^{ab}

Stages[A]: Sanitation stage; Stage 0: Immediately after the completion of shift work; Stage 1: After debris removal by dry cleaning and washing with pressurized warm water, and right before the application of the detergent; Stage 2: After the belt surfaces were cleaned with detergent and thoroughly rinsed, and right before the application of sanitizer; sanitizer application; Stage 3: within 5 min of the application of sanitizers; [B]N: Number of samples from which no aerobes were recovered.

NA: Not applicable as no drain samples were collected near Belt-1.

Data are mean ± standard deviations ($n = 12$ for belt and drain; $n = 6$ for air). Counts for belt surfaces, drain, and air were compared among their respective sanitation stages and between belts where applicable. Means with different letters are significantly different ($P < 0.05$).

Table 2
Enterobacteriaceae and coliforms recovered from conveyor belts and drains (n = 12) in a beef fabrication facility

Surface	Stages[A]	*Enterobacteriaceae*			Coliforms		
		N (b1/b2)[b]	Belt-1	Belt-2	N(b1/b2)	Belt-1	Belt-2
Belt	Stage 0	4/12	1.79 ± 1.09[b]	ND[c]	6/8	1.38 ± 1.01[b]	0.96 ± 0.70[ab]
	Stage 1	0/0	3.32 ± 0.42[c]	3.47 ± 0.41[c]	0/0	2.68 ± 0.37[c]	2.85 ± 0.41[c]
	Stage 2	11/4	0.57 ± 0.24[a]	1.24 ± 0.76[ab]	12/10	ND	0.70 ± 0.53[ab]
	Stage 3	9/10	0.66 ± 0.29[a]	0.86 ± 1.04[a]	12/9	ND	0.84 ± 0.81[ab]
Drain	Stage 0	0	NA	5.18 ± 0.65[b]	0	NA	4.83 ± 0.71[b]
	Stage 1	0	NA	6.50 ± 1.34[c]	0	NA	5.08 ± 1.10[b]
	Stage 2	0	NA	3.40 ± 1.10[a]	2	NA	2.81 ± 1.34[a]
	Stage 3	0	NA	3.23 ± 0.95[a]	2	NA	2.83 ± 1.40[a]

Stages[B]: Sanitation stage; Stage 0: Immediately after the completion of shift work; Stage 1: After debris removal by dry cleaning and washing with pressurized warm water, and right before the application of the detergent; Stage 2: After the belt surfaces were cleaned with detergent and thoroughly rinsed, and right before the application of sanitizer; sanitizer application; Stage 3: within 5 min of the application of sanitizers; [B]N: the number of samples from which no *Enterobacteriaceae* or coliforms were recovered from belt-1 (b1) or belt-2 (b2); [C]ND: none of the samples yielded counts for that particular bacterial group.
NA: Not applicable as no drain samples were collected near belt-1.
Data are shown as means ± standard deviation (n = 12). Counts ($\log_{10}$ CFU/1000 cm^2) for belt surfaces and drain were compared among their respective sanitation stages and between belts where applicable. Means with different letters are significantly different ($P < 0.05$).

Table 3
Escherichia coli recovered from conveyor belts and drains (n = 12) in a beef fabrication facility

Sample	Sanitation		*E. coli* (Log_{10} CFU/1000 cm^2)	
Groups	Stages[A]	N(b1/b2)[b]	Belt-1	Belt-2
Belt surfaces	Stage 0	0/0	ND[c]	ND
	Stage 1	4/12	0.96 ± 0.69	1.68 ± 0.33
	Stage 2	0/1	ND	0.55 ± 0.19
	Stage 3	0/0	ND	ND
Drain	Stage 0	3	NA	1.20 ± 1.27
	Stage 1	5	NA	1.24 ± 1.04
	Stage 2	5	NA	1.12 ± 0.83
	Stage 3	2	NA	1.01 ± 1.20

Stages[A]: Sanitation stage; Stage 0: Immediately after the completion of shift work; Stage 1: After debris removal by dry cleaning and washing with pressurized warm water, and right before the application of the detergent; Stage 2: After the belt surfaces were cleaned with detergent and thoroughly rinsed, and right before the application of sanitizer; sanitizer application; Stage 3: within 5 min of the application of sanitizers; [B]N: the number of samples from which *E. coli* were recovered from belt-1 (b1) or belt-2 (b2); [C]ND: none of the samples yielded counts for *E. coli*.
NA: not applicable as no drain samples were collected near belt-1.
Data are shown as means ± standard deviations (n = 12).

bacterial groups throughout sanitation largely followed a similar trend to the total aerobic count with the numbers at postsanitization being 2 log lower than those prior to cleaning. No *Enterobacteriaceae*, coliforms, or *E. coli* were recovered from any of the air samples.

Overall, 17 of the 96 samples from the belts yielded *E. coli* count, 16 of which were from samples collected at Stage 1 (Table 3). Mean values for *E. coli* were 1.0 and 1.7 log CFU/1000 cm^2 for belts 1 and 2 at this stage. No *E. coli* were recovered from any of the samples from the belts before cleaning or after sanitation. For the drain samples, less than 50% of the samples yielded *E. coli* at any given sanitation stage, with counts not being affected by sanitation stages.

Bacterial communities of air, belt-surface, and drain during sanitation as determined by 16S rRNA gene amplicon sequencing

Differences between the air, belt, and drain bacterial communities were observed. Alpha-diversity indices indicated that ASV estimates in air and drain samples were higher ($P < 0.05$) in richness and evenness compared to those from belts (Fig. S1). Beta-diversity analyses identified differences in the microbial communities between the air, belts, and drains ($P < 0.01$), with a core shared by all three sources (Fig. 2a). Stage 1 notably altered the air microbial community (Fig. 2b1; $R^2 = 0.11$, $P < 0.05$), with an increase in microbial taxa compared to Stage 0 (Fig. S2a&d, $P < 0.05$). Regarding belts, the microbiome at Stage 0 differed from Stages 1–3 of the sanitation process (Fig. 2b2; $R^2 = 0.10$, $P < 0.01$) and at Stage 1 differed from Stage 3 ($P < 0.05$). Additionally, Stage 1 had higher taxon-richness compared to other stages (Fig. S2b; $P < 0.05$), whereas taxon evenness was comparable among the stages. For the drain samples, all paired comparisons except for Stage 2 and Stage 3 differed (Fig. 2b3, $P \leq 0.01$). However, no differences ($P > 0.05$) among the stages were observed for alpha-diversity (Fig. S2c&f).

The microbial composition varied by sampling day (trip) for belt, air, and drain samples (Fig. 3). *Acinetobacter* was predominant in all three sample types, with an overall relative abundance of 51.18%, 46.06%, and 55.83% for air, belt, and drain, respectively. Additionally, several other genera, including *Enhydrobacter*, *Methylobacterium*, *Chryseobacterium*, *Klebsiella*, and *Pseudomonas,* were also commonly observed, with an overall relative abundance ranging from 1.16% to 9.06%. When profiling the microbial communities of the original air samples and enriched air samples, a total of 23 genera exceeded 1% of relative abundance in both. Original air samples contained 22 genera, while enriched samples yielded 21 genera (Fig. S3). Most genera (20/22) that represented more than 1% of the microbial population in the original samples were also found above this threshold in the enriched samples. However, *Klebsiella* and *Thermus*, present in the original air samples, were not detected following enrichment. Interestingly, *Microbacterium* was undetectable in the original air samples, but was present in enriched air samples.

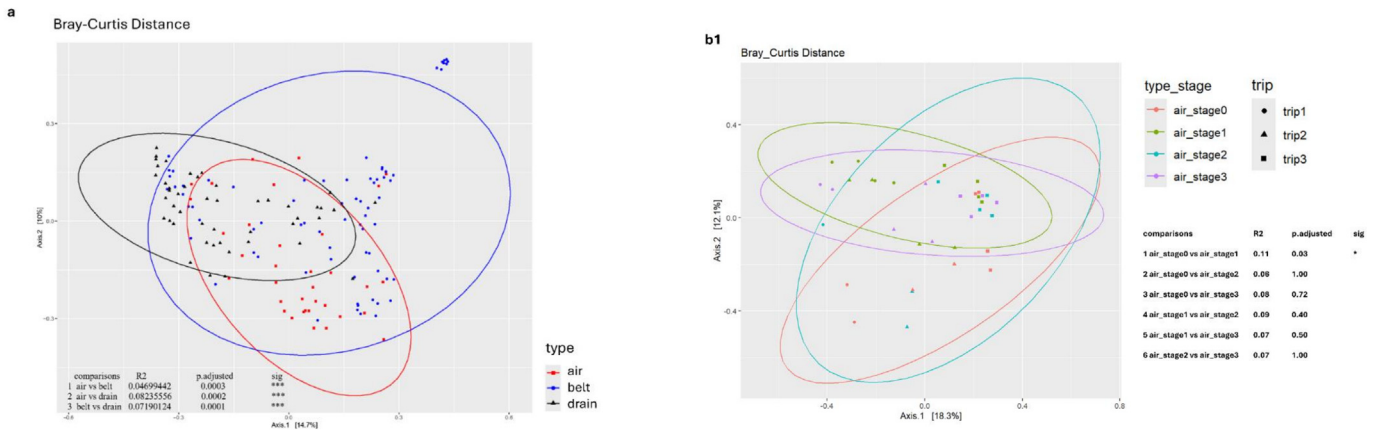


Figure 2. Principal coordinate analysis of the Bray-Curtis dissimilarities of the overall microbiota of air, belts, and drain (a), and as separated by sanitation stages for the individual groups of air (b1), belt surface (b2), and drain (b3). Stages: Sanitation stage; Stage 0: Immediately after the completion of shift work; Stage 1: After debris removal by dry cleaning and washing with pressurized warm water, and right before the application of the detergent; Stage 2: After the belt surfaces were cleaned with detergent and thoroughly rinsed, and right before the application of sanitizer; sanitizer application; Stage 3: within 5 min of the application of sanitizers.

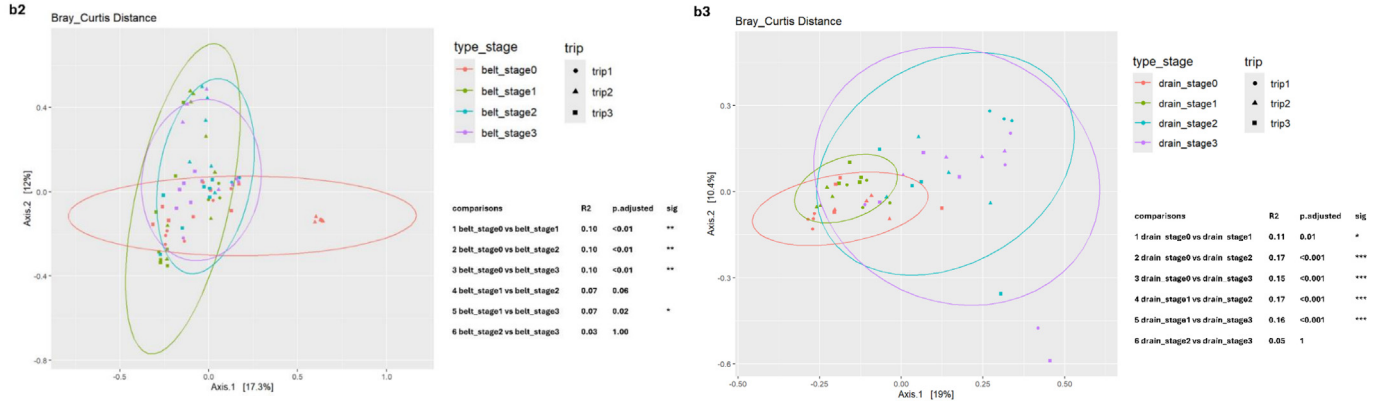


Fig. 2 (*continued*)

Dynamics of the processing plant microbial communities during sanitation

To better understand how the microbial communities shifted during the sanitation stages, differentially abundant taxa comprising over 5% of the respective microbiomes of air, belt, and drain samples were determined by MaAslin2 with adjusted *P* value ($P < 0.05$). For air (Fig. 4a), the relative abundance of *Thermus* and *Klebsiella* was 6.15% ± 4.30 and 6.78% ± 3.17 at Stage 0. The respective numbers decreased to 0.79% ± 0.50 and 1.8% ± 0.91 at Stage 1 and remained low throughout the sanitation process. On the other hand, *Sphingomonas* was enriched as sanitation progressed, from not being detected to a relative abundance of up to 5.65% ± 1.23 at Stage 3. The mean relative abundance of *Acinetobacter* remained stable in the air microbial community during sanitation stages, ranging from 45.67% ± 4.78 to 47.12% ± 9.85. For belts (Fig. 4b), the relative abundance of *Pseudomonas* was effectively decreased by Stage 1, dropping from 20.89% ± 4.70 to 4.20% ± 0.93, and remained at approximately 3%. In contrast, *Enhydrobacter* (4.97% ± 2.01 at Stage 0) and *Methylobacterium* (1.06% ± 0.50 at Stage 0) progressively increased in relative abundance throughout sanitation, and peaked at Stage 3 (14.70% ± 5.94 and 9.62% ± 2.8 for *Enhydrobacter* and *Methylobacterium*, respectively). For *Acinetobacter*, its relative abundance increased following Stage 1 cleaning, from 46.18% ± 4.77 to 62.60% ± 6.02 (Stage 0 vs. Stage 1, $P = 0.12$). After detergent application in Stage 2, the relative abundance of *Acinetobacter* decreased to 32.96% ± 6.48 (Stage 1 vs. Stage 2, $P = 0.09$) and continued to decline to 25.98% ± 5.11 after the final sanitization (Stage 2 vs Stage 3, $P = 0.35$). Notably, by the end of sanitation, the relative abundance of *Acinetobacter* on belts was lower compared to that before cleaning (Stage 0 vs. Stage 3, $P = 0.05$). Significant compositional shifts in microbial communities were also observed in drain samples (Fig. 4c). *Acinetobacter* notably decreased after Stage 1, from 68.28% ± 3.33 at Stage 0 to 41.82% ± 7.09 at Stage 3. The relative abundance of *Pseudomonas* and *Stenotrophomonas* shifted during sanitation, but eventually returned to levels comparable to those at Stage 0.

The origin of the microbes on belts during the sanitation process was estimated using SourceTracker2. Designating the belt at each stage as the target environment, it was predicted that at Stage 0, 57.4% of microbes on the belt originated from the drain, which decreased to 42.2% at Stage 1 (Fig. 5a). As sanitation progressed, air replaced the drain, becoming a major contributing source to the belt microbial community. Before cleaning started, 15.8% of the belt-surface microbial community was predicted to have originated from the air, which changed to 38.7% at Stage 1, 40.9% at Stage 2, and 43.4% at Stage 3 (Fig. 5a). When air, belt surface, and drain were each considered as a target environment, it was predicted that over 40% of the drain microbial community originated from the belt surface in Stage 1 (Fig. 5b). For the belt surface microbial community, the top three sources were Stage 0 drain at 14.1%, Stage 2 air at 11.45%, and Stage 1 air at 9.10%.

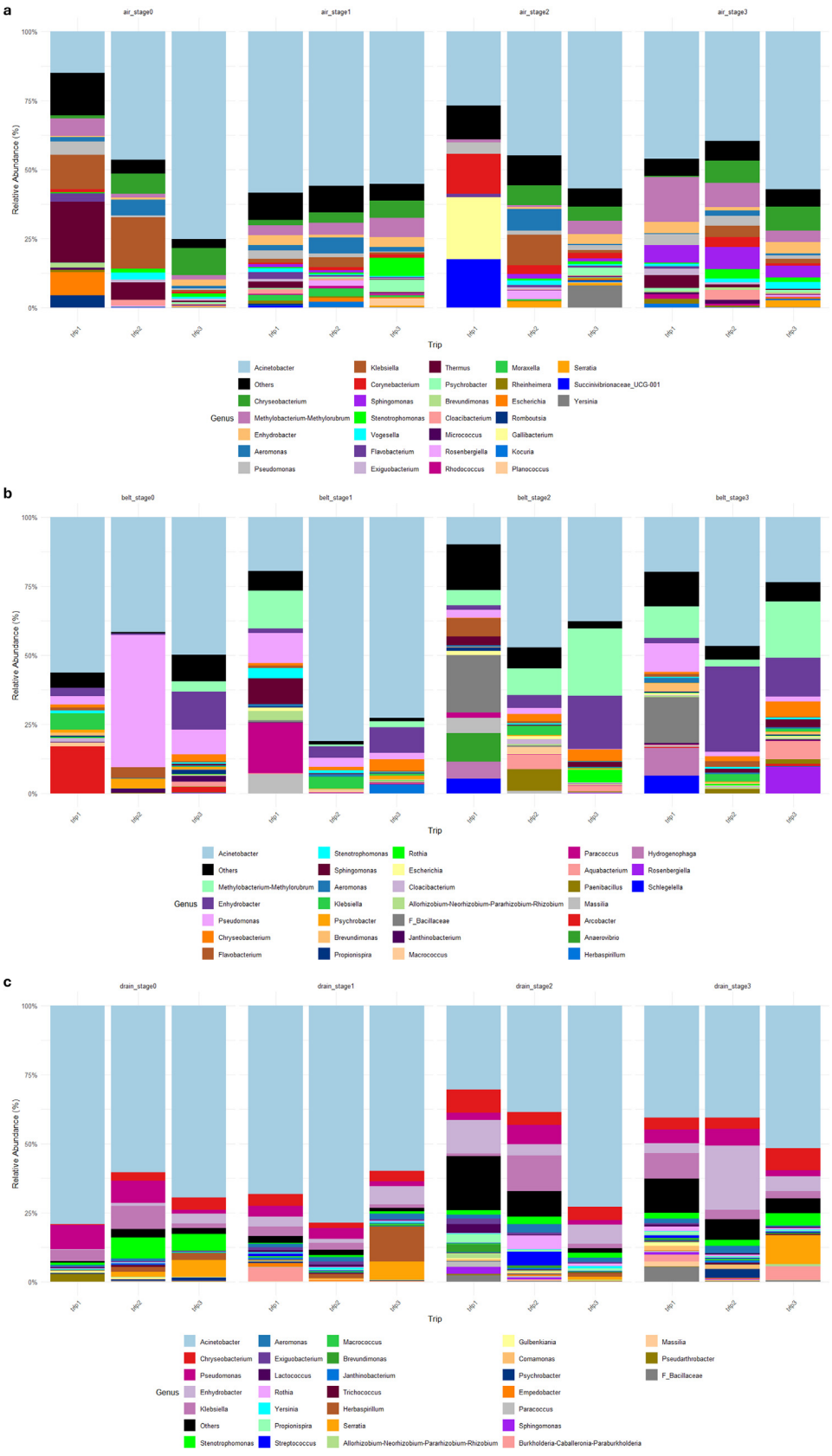


Figure 3. Mean relative abundance of top 30 bacterial genera of the air-, belt surface-, and drain-associated microbiota. Stacked bar plot showing the mean relative abundance (%) of bacterial genera on the same sample type and cleaning stage in each sampling trip. Stages: Sanitation stage; Stage 0: Immediately after the completion of shift work; Stage 1: After debris removal by dry cleaning and washing with pressurized warm water, and right before the application of the detergent; Stage 2: After the belt surfaces were cleaned with detergent and thoroughly rinsed, and right before the application of sanitizer; sanitizer application; Stage 3: within 5 min of the application of sanitizers.

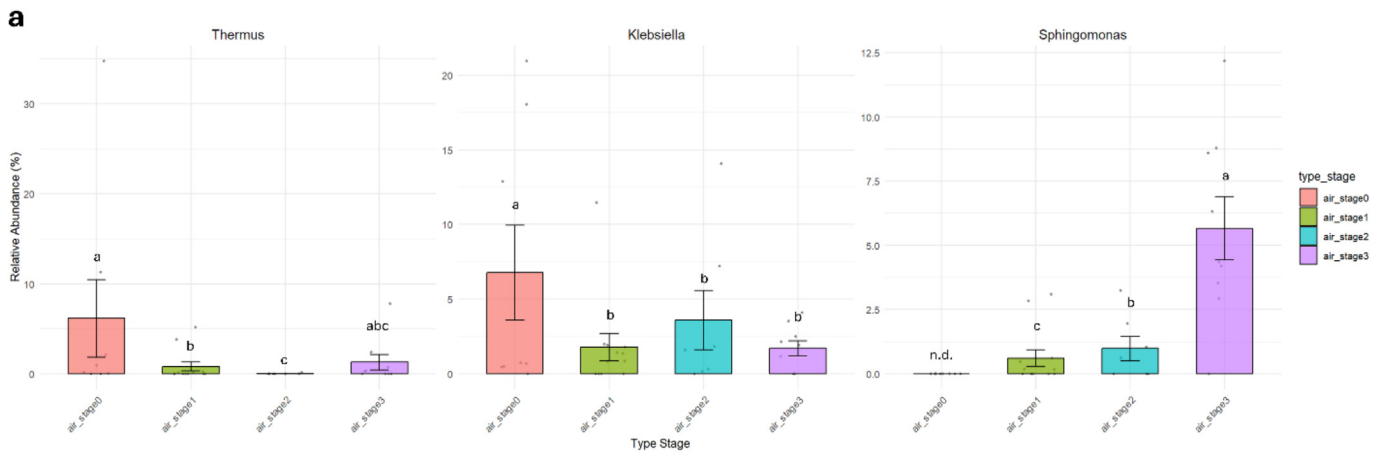


Figure 4. Differentially abundant taxa between cleaning stages in the air microbiota (a), belt-surface (b), and drain (c) as determined by MaAsLin2. Bar plots show mean relative abundance ± standard error of mean. Significantly differentially abundant taxa (FDR-adjusted $p \leq 0.05$) with mean relative abundance over 5% are displayed. Stages: Sanitation stage; Stage 0: Immediately after the completion of shift work; Stage 1: After debris removal by dry cleaning and washing with pressurized warm water, and right before the application of the detergent; Stage 2: After the belt surfaces were cleaned with detergent and thoroughly rinsed, and right before the application of sanitizer; sanitizer application; Stage 3: within 5 min of the application of sanitizers.

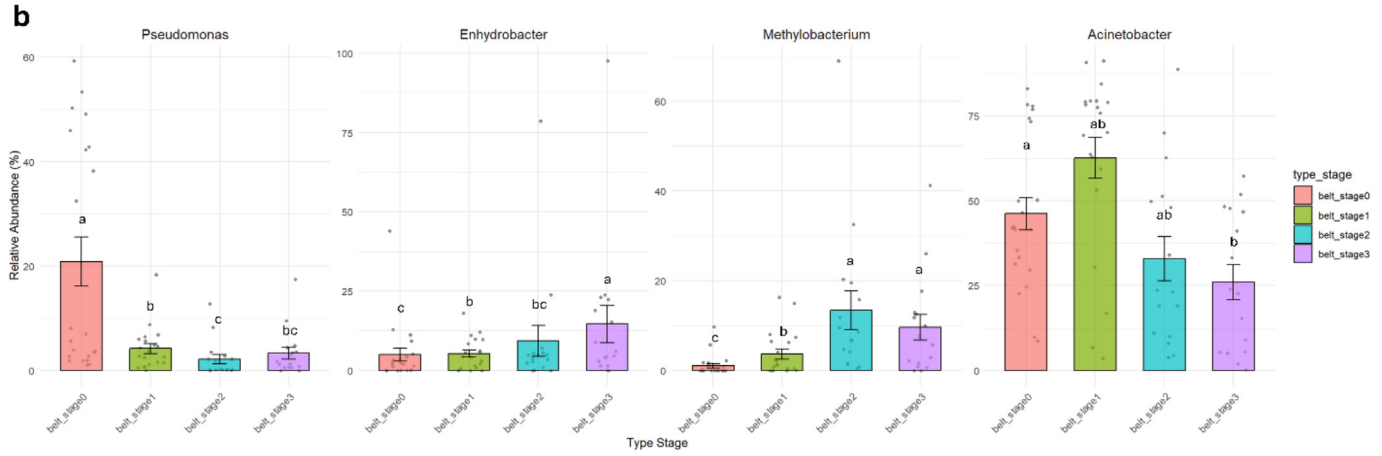


Fig. 4 (*continued*)

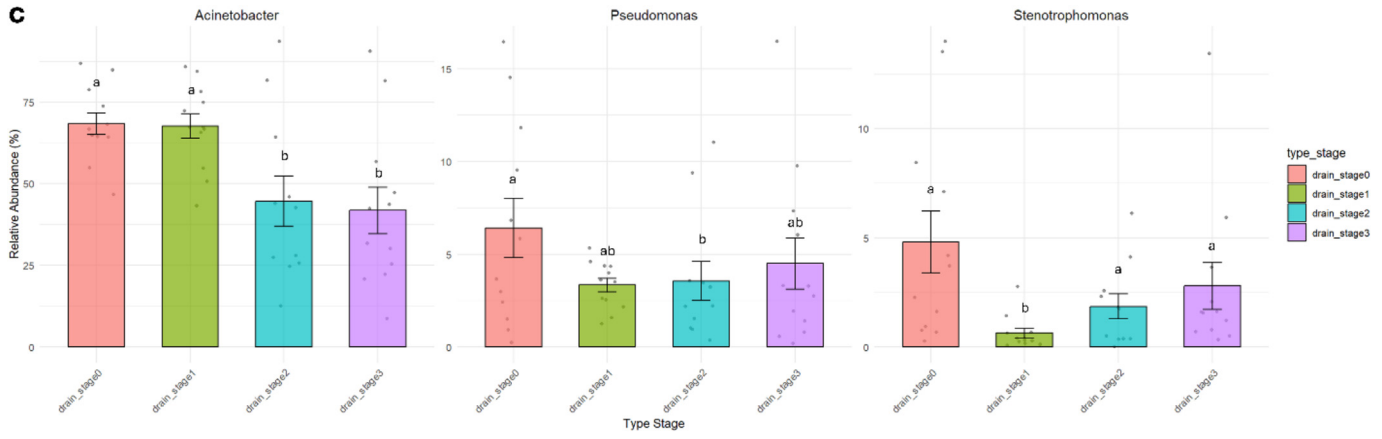
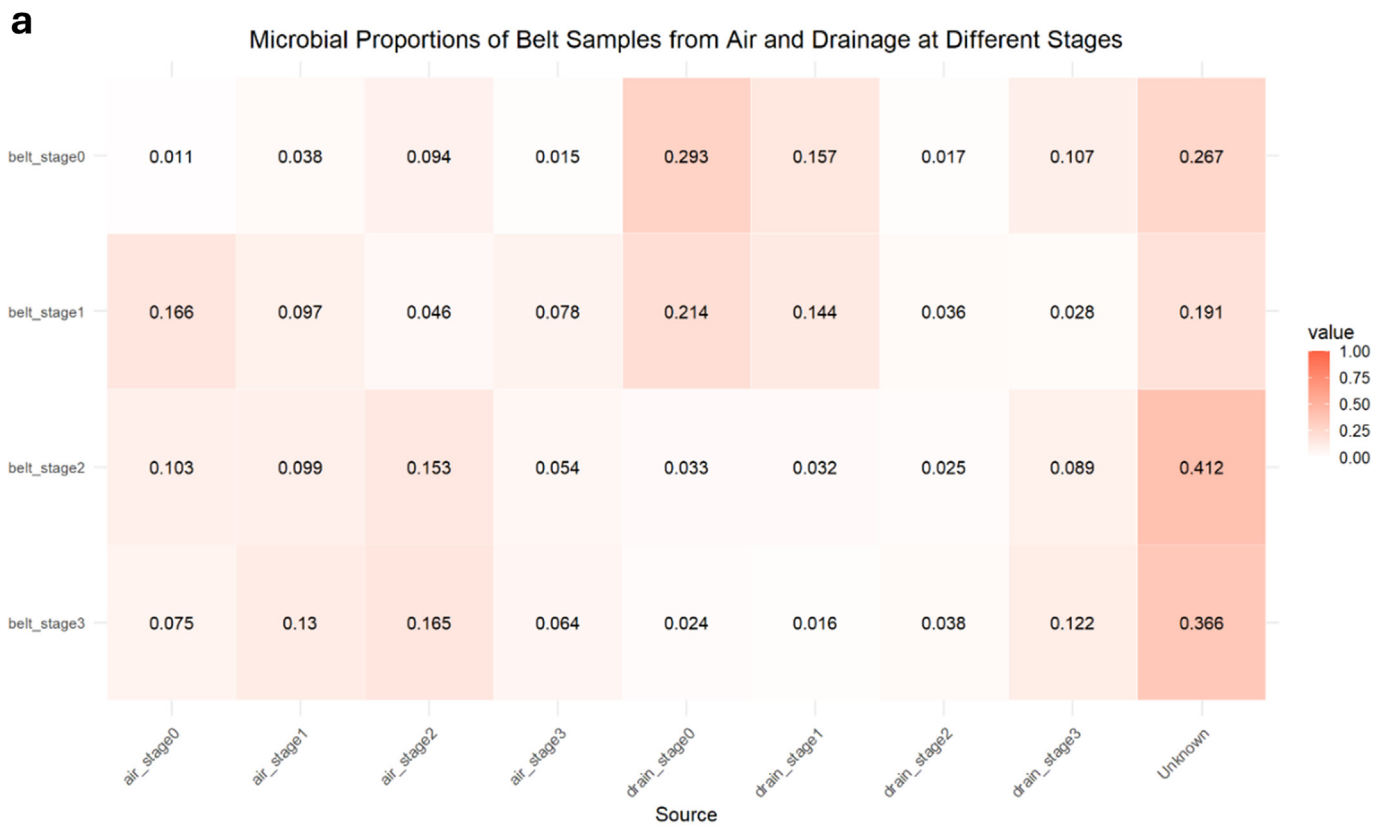


Fig. 4 (*continued*)



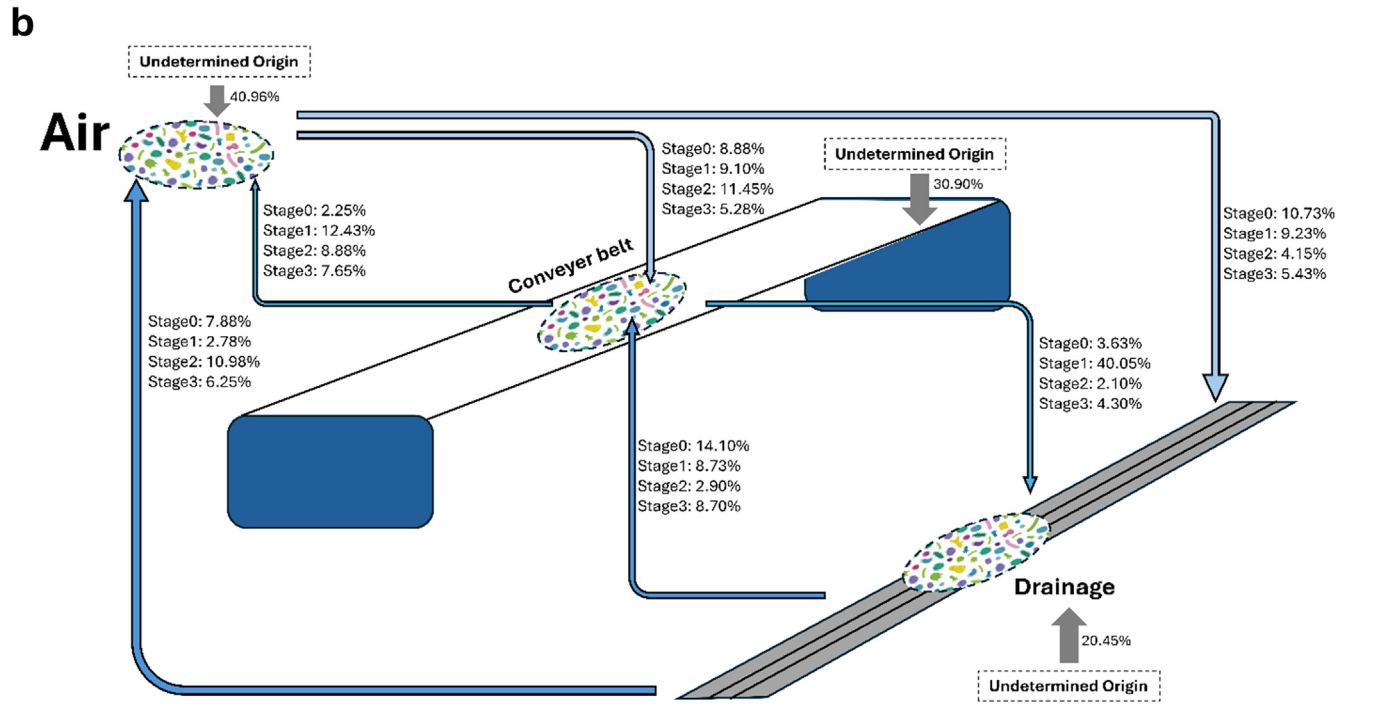


Figure 5. Heatmap of the estimated proportions of microbes on belt surfaces originating from different sources across sanitation stages determined by SourceTracker (a) and a proposed pathway for translocation of bacteria during cleaning and sanitation (b). Stages: Sanitation stage; Stage 0: Immediately after the completion of shift work; Stage 1: After debris removal by dry cleaning and washing with pressurized warm water, and right before the application of the detergent; Stage 2: After the belt surfaces were cleaned with detergent and thoroughly rinsed, and right before the application of sanitizer; sanitizer application; Stage 3: within 5 min of the application of sanitizers.

Discussion

The overall microbial load on conveyor belts in the fabrication facility involved in this study, prior to the start of the sanitation process (Stage 0), aligned with previous studies (Wang et al., 2018; Youssef et al., 2013). In these studies, aerobes, *Enterobacteriaceae*, coliforms, and *E. coli* on the conveyor belt surfaces were in the range of 2.5 – 6, < 3, and <2.5, <0.5 log CFU/ 1000 cm^2, respectively. The total aerobic counts and enteric organisms reported in this study were comparable to the lowest numbers reported for conveyor belts at this stage. In contrast to our study, which found aerobic counts in air at approximately 2.5 log CFU/ 1000 L, a previous study by Schmidt and coworkers showed that the number of total aerobes in most air samples collected from beef fabrication facilities was below the detection limit (Schmidt et al., 2012). This significant difference may reflect differences in air sampling methods. The study by Schmidt and coworkers used a moist sponge and impaction-based methods to collect air samples, whereas the present study employed a novel method that collected air samples on filter fibers, on which bacterial cells were deposited and electrostatically retained (Boifot et al., 2020). Irrespective of sanitation stages, *Enterobacteriaceae*, coliforms, and *E. coli* in the air were below the detection limit of 1 log CFU/100 L in the present study. To date, to the best of our knowledge, very few studies have addressed the microbial load in the air of meat fabrication facilities during the sanitation process. Previously, Reiche and coworkers reported microbial reduction ranging from 1.4 to 2 log CFU/cm^2 on drain surfaces at a salmon processing facility after cleaning and disinfection (Reiche et al., 2024). However, there remains a paucity of information on the microbial load of drain surfaces in beef fabrication facilities, irrespective of sanitation stages and as such, making it challenging to compare the findings of the present work to published accounts.

In the present study, the significant increase in bacterial load at Stage 1 compared to Stage 0 on conveyor belts supports the hypothesis of bacterial translocation to belt surfaces. Similarly, Wang and coworkers reported a trend of an increase in aerobic bacteria on the conveyor belt after debris removal in a beef fabrication facility, although the observed increase was not significant (Wang et al., 2018). This increase in bacterial counts could have resulted from the redistribution of bacteria in loosely attached, attached, and/or sessile form from different areas on the belts by water. The numbers of bacteria present on nonfood contact surfaces in general are higher than those for food-contact surfaces of the belts (Wang et al., 2018; Yang, Wang, et al., 2023). However, the increase in bacterial counts in air at Stage 1 in the present study, corroborated by the source attribution model, suggests that water droplets containing bacteria/bioaerosols generated by the pressurized water washing of floors would be another contributing factor. The extent of dislodging bacteria from floor surfaces to air and subsequent deposition would depend on the water pressure as well as temperature. The reduction in bacterial counts by foaming and degreasing (Stage 2) and a lack of further reduction by sanitization (Stage 3) were in line with previous studies (Wang et al., 2018; Yang et al., 2017). A consistent reduction of more than 2 log in total aerobes, *E. coli*, coliform, and *Enterobacteriaceae* after Stage 2 (detergent washing and rinsing) compared to Stage 1 was observed for both drains and belts. This substantial decrease highlights the effectiveness of the detergent-based cleaning step in removing bacteria. This, together with no difference in the microbial community at Stage 2 and Stage 3, suggests that the effect of subsequent sanitizer is limited. This finding is notable considering that the plant uses neutral quat, lactic acid quat, and peracetic acid in rotation. Al-Qadiri and coworkers investigated the effect of quat at 200 ppm manually sprayed on polypropylene cutting boards that had been artificially contaminated with five bacterial strains, *Salmonella* Typhimurium, *E. coli* O157:H7, *Listeria monocytogenes*, *Campylobacter jejuni,* and *Staphylococcus aureus* at 7–8 log CFU/100 cm^2 (Al-Qadiri et al., 2016). They found that all five bacterial strains were reduced by > 5 log after 5 − min exposure at ambient temperature (22 °C) (Al-Qadiri et al., 2016). To qualify for a food-contact surface sanitizer, a 5-log reduction of one or two bacterial strains with 30 s of exposure is required (Health Canada, 2025). On the other hand, no difference was found in the total aerobic count or the lactic acid bacteria population on the food-contact surface of a conveyor belt at a beef processing plant between after cleaning with detergent and after application of sanitizer (Wang et al., 2018). In that study, the total aerobic counts for conveyor belts prior to sanitation were between 1 and 2.6 log CFU/cm^2. Reiche and coworkers investigated the microbial efficacy of cleaning and disinfection at a salmon processing plant and found an average reduction of 1.1 log CFU/cm^2 by the entire process (Reiche et al., 2024). However, the reduction varied with season, the type of surface, and the location of the same type of surface. For example, conveyor belts at three locations had no change in microbial counts in summer as a result of the cleaning and disinfection process, but had an overall reduction >1 log in winter. For winter, the reduction of total aerobic counts for the three belts varied from 0.1 to 1.8 log CFU/cm^2. The initial bacterial load for belts was mostly between 2 and 3 log CFU/cm^2. The difference between the efficacy of the sanitization step for the conveyor belts in commercial food processing facilities and the expected efficacy based on lab studies could be due to multiple factors, such as bacterial species/strains, level of contamination, type and condition of surface, and the conditions under which the target bacteria encountered prior to the treatment. In addition, conveyor belts are commonly kept on running during the application of sanitizer, which may bring in an influx of bacteria, and there would be residual water from the previous rinsing step, compared to static laboratory conditions.

The finding that the majority of microbial communities in air, belt, and drain samples were predominantly Proteobacteria, Actinobacteria, Firmicutes, and Bacteroidetes (~90%) is in agreement with a previous report on bacteria present on meat processing surfaces, regardless of sanitation stages (Botta et al., 2020). Our results show that the microbial community on conveyor belts and in drains and, to a lesser extent, air was primarily Gram-negative, psychrotrophs, with *Acinetobacter* being most abundant in all samples. Møretrø and Langsrud (Møretrø & Langsrud, 2017) compiled studies investigating bacteria on postsanitation surfaces in the food industry and found that Gram-negative bacteria that could grow at low temperatures dominated high moisture environments such as meat and fish processing facilities, with members of *Pseudomonas* being the most prevalent bacterial genus. Recently, a large-scale metagenomic study of 19 facilities for mostly processed meat has reported that *Acinetobacter* was among the top five genera found in raw materials, processing environments, and final products, along with *Pseudomonas*, *Staphylococcus*, *Brochothrix*, and *Psychrobacter* (Barcenilla et al., 2024). *Pseudomonas*, *Psychrobacter*, and *Acinetobacter* have been reported to be predominant on postsanitation surfaces at fresh meat processing facilities as well (Yang, Narvaez-Bravo, et al., 2023). A shotgun-metagenomic study investigating the microbiome of floor-drains at various locations (Hotscale, Hotbox, Cooler, Processing, and Grind room) in three beef processing facilities over two time points reported that *Pseudomonas*, *Psychrobacter*, and *Acinetobacter* were the top three genera across location, plant, and time point (Palanisamy et al., 2023). Even so, the prevalence for specific taxa varied with locations and plants. The prevalence of these genera may reflect their ability to survive/thrive under such environments and the variation would reflect the dynamic nature of bacteria in such environments (Møretrø et al., 2013; Yuan et al., 2018). Different from previous reports, in addition to *Acinetobacter* and *Pseudomonas*, the present study also found sizable fractions of *Enhydrobacter* and *Methylobacterium* in belt and drain samples. They have been reported to be predominant taxa in cloud water (Amato et al., 2017), air-conditioning units (Watanabe et al., 2022), and water-heating systems

(Kjellerup et al., 2005). However, despite their prevalence, the specific ecological role of those bacterial taxa in the overall microbial community and functionality in meat processing environments remains unclear.

After Stage 1 sanitation and in subsequent stages, a notable decrease in the relative abundance of *Pseudomonas* on the conveyor belt was observed in the present study. There were no biocides applied other than pressurized water wash and the running of the belts at Stage 1. Therefore, it is reasonable to speculate that this decrease would likely reflect the change in microbial composition as a result of dislodging of bacterial cells from non-contacting surfaces such as the belt and floor/drain, rather than an absolute reduction in the number of *Pseudomonas*. The increase of total aerobes in air and belt samples may be a consequence of mechanical dislodgement and aerosolization followed by redeposition on the belt surfaces of some surface floor/wall adhered bacteria by the pressurized water wash. Similarly, the trend of increased *Acinetobacter* on the conveyor belt postpressurized water wash may also result from water droplet transporting the dislodged *Acinetobacter* from the ambient environment during pressurized wash and redepositing them onto the belt surfaces. The decrease in the relative abundance of *Thermus* and *Klebsiella* in the air after Stage 0 was likely a result of aerosolization of other bacterial species. Interestingly, during cleaning, we observed a gradual increase in the composition of some bacterial genera in the air and belt surface that were previously reported as residents in air- and/or water-systems (Gulati & Ghosh, 2017; Satari et al., 2023) including *Sphingomonas, Enhydrobacter,* and *Methylobacterium*. The observed increase in relative abundance of these genera was likely attributed to the displacement of *Acinetobacter*. The water used during cleaning as well as aerosolization during pressure washing would result in mobilization of bacteria at different locations. Indeed, permutating the origin and destination of bacteria, our bacterial translocation model indicated a high level of bacteria transmission between air, belt surface, and the drain.

Even though both culture−based method and microbial profiling were used in the present study to discern the dynamics of the microbial communities in the beef fabrication facility, there are some limitations to these methods. Culture-based methods are limited in their ability to recover bacteria that are fastidious, while the sequencing method does not differentiate DNA from dead and viable bacteria and is affected by sequencing bias. The comparison between taxa in the original air samples and their respective enrichment cultures would suggest that the 16S profiling does serve as a reasonable representation of the microbial diversity of the viable fraction of the air microbial community. Any sampling at large facilities also provides only a glimpse of the entirety of the environment with respect to microbial contaminants. Even so, the findings of the present study clearly demonstrated that the final microbiota on food-contact surfaces is influenced by the application of biocides and translocation of bacteria from other sources. Specifically, the final microbiota present on food-contact surfaces, such as conveyor belts, appear to result not only from the selective pressure of biocides but also from bacterial translocation during sanitation, particularly through the application of pressurized water. The sizable relative abundance and prevalence of *Acinetobacter, Pseudomonas, Enhydrobacter, Methylobacterium, and Stenotrophomonas* on postsanitation surfaces would warrant further examination of their response to sanitation and their role in such an ecological niche. Our results suggest that the current approach to sanitizer application, especially at the final stage of sanitation, may be insufficient to prevent cross-contamination and warrants reassessment to improve overall hygiene and food safety outcomes.

CRediT authorship contribution statement

Barun Yadav: Writing – original draft, Methodology, Formal analysis, Data curation. **Yi Fan:** Writing – original draft, Methodology, Formal analysis. **Scott Hrycauk:** Methodology, Investigation. **Tim McAllister:** Writing – review & editing, Funding acquisition. **Claudia Narvaez-Bravo:** Writing – review & editing, Funding acquisition. **Tyson Brown:** Writing – review & editing, Funding acquisition. **Xianqin Yang:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary material to this article can be found online at https://doi.org/10.1016/j.jfp.2025.100647.

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